

EXPERIMENT – 28

Implementation of Modified Diffie-Hellman Key Exchange Protocol

CODE:

```
#include <stdio.h>

int main()
{
    int a = 5;          // Public number
    int x = 3, y = 7;   // Alice and Bob secret numbers
    int A, B, key;

    // Alice sends x * a
    A = x * a;

    // Bob sends y * a
    B = y * a;

    printf("Public number a = %d\n", a);
    printf("Alice secret x = %d\n", x);
    printf("Bob secret y = %d\n", y);

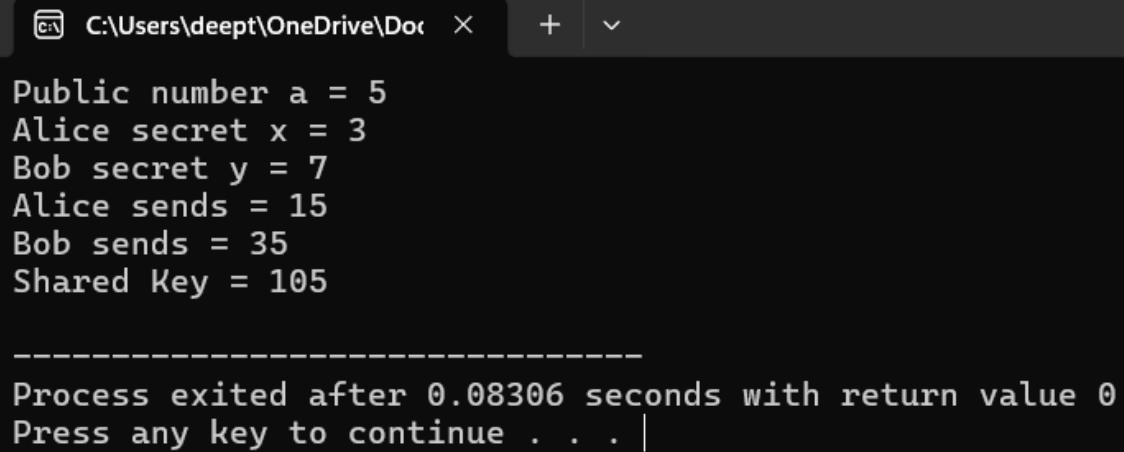
    printf("Alice sends = %d\n", A);
    printf("Bob sends = %d\n", B);

    // Both calculate the same key
    key = (A * B) / a;
```

```
printf("Shared Key = %d\n", key);

return 0;
}
```

OUTPUT:



A screenshot of a Windows command prompt window. The title bar shows the file path "C:\Users\deept\OneDrive\Doc" and standard window controls. The command prompt displays the following output:

```
Public number a = 5
Alice secret x = 3
Bob secret y = 7
Alice sends = 15
Bob sends = 35
Shared Key = 105

-----
Process exited after 0.08306 seconds with return value 0
Press any key to continue . . . |
```